

Aninag: Barangay Transparency Fund Management System with Blockchain-Based Financial Record Verification

Amiel Carl A. Santos

National University, Philippines

I. PROPOSED TITLE

Aninag: Barangay Transparency Fund Management System with Blockchain-Based Financial Record Verification

II. BACKGROUND OF THE STUDY

Barangays serve as the foundation of local governance in the Philippines, acting as the first point of contact between citizens and government services. They are responsible for delivering basic public services, implementing community development programs, maintaining peace and order, managing public resources, and ensuring that government funds are utilized for the benefit of the community. As communities continue to grow, barangay offices face increasing challenges in managing financial records, monitoring expenditures, preparing financial reports, and maintaining transparency while ensuring that public funds are used efficiently and responsibly.

Barangay Ulingao is one of the barangays located in the Municipality of San Rafael, Province of Bulacan, under Region III (Central Luzon). It is a land area community with a population of approximately 3,711 residents, representing a significant portion of the municipality's population. As the barangay continues to develop and implement community projects, infrastructure improvements, and public services, the demand for efficient financial management and transparent reporting also increases. This highlights the need for a digital system that supports accurate financial record management while allowing residents to remain informed about the utilization of public funds.

Traditionally, many barangays continue to rely on manual record-keeping, paper-based documentation, and spreadsheet applications for recording financial transactions, monitoring budgets, and preparing financial reports. Although these methods have long been used in local government operations, they may result in delayed report generation, difficulty in retrieving financial records, increased risk of human error, misplaced documents, and limited transparency regarding how barangay funds are allocated and spent. As financial transactions increase over time, manual processes become more difficult to manage and may affect the efficiency and accountability of barangay financial operations.

One of the important responsibilities of barangay officials is the proper management of public funds allocated for infrastructure projects, community programs,

emergency response, administrative expenses, and other government services. Residents, however, often have limited access to financial information and may only obtain updates through periodic reports or public meetings. This lack of accessible financial information may reduce public awareness, limit community participation, and affect public confidence in the management of barangay resources.

Another concern is the authenticity and integrity of official financial reports and documents. Budget reports, disbursement records, liquidation reports, and other financial documents are commonly verified through manual inspection, which may require additional time and effort from both barangay officials and auditing authorities. Without a secure verification mechanism, financial documents may become vulnerable to unauthorized modifications, duplication, or fraudulent use, potentially affecting the credibility of financial records and public trust.

With the advancement of information technology, web and mobile-based information systems provide opportunities to improve transparency, accessibility, and efficiency in local government financial management. A centralized digital platform allows authorized barangay officials to record financial transactions, monitor budgets, generate reports, and manage financial records more efficiently. At the same time, residents can conveniently access approved financial reports, monitor barangay fund utilization, view project expenditures, and receive updates through a mobile application, promoting greater transparency and community participation.

Furthermore, blockchain technology provides an additional layer of security by creating tamper-evident records through cryptographic hashing. Instead of storing confidential financial documents directly on the blockchain, the system stores only their unique cryptographic hashes, allowing financial records to be verified while preserving data privacy. Through QR code-based verification, residents, auditors, and authorized government agencies can quickly verify the authenticity of official financial reports and determine whether a document has been altered after its issuance.

Therefore, this study proposes Aninag: A Web and Mobile-Based Barangay Transparency Fund Management System with Blockchain-Based Financial Record Verification. The proposed system aims to provide Barangay Ulingao, San Rafael, Bulacan with a secure, transparent, and accessible platform for managing barangay funds, monitoring financial transactions, generating transparency reports, and verifying the authenticity of official financial documents. By integrating web and mobile technologies with blockchain-based verification, the proposed system seeks to improve financial accountability, strengthen public trust, discourage corruption, and support transparent governance within the barangay.

III. PROBLEM STATEMENT

General Problem

How can a web and mobile-based barangay transparency fund management system improve financial transparency, accountability, and public trust through blockchain-based financial record verification?

Specific Problems

1. How can barangay financial transactions be digitally recorded and managed more efficiently?
2. How can residents conveniently monitor approved barangay funds, expenditures, and project budgets through a mobile application?
3. How can blockchain technology be used to verify the authenticity and integrity of financial records?
4. How can QR code verification simplify the validation of official financial reports and documents?
5. How can reports, analytics, and audit logs improve accountability and financial monitoring among barangay officials?

IV. OBJECTIVES

General Objective

To develop a web and mobile-based barangay transparency fund management system with blockchain-based financial record verification that promotes transparency, accountability, and public trust in barangay financial management.

Specific Objectives

1. To develop a web-based administrative system for recording and managing barangay financial transactions, budgets, and expenditures.
2. To develop a mobile application that allows residents to monitor approved financial reports, project expenditures, and budget utilization.
3. To implement blockchain-based verification by generating cryptographic hashes for finalized financial records.
4. To integrate QR code verification for validating the authenticity of official financial reports and documents.
5. To provide reports, analytics, and audit logs that support financial monitoring and decision-making.

V. TARGET USERS

Primary Users

Barangay Officials

- Barangay Captain
- Barangay Treasurer
- Barangay Secretary
- Authorized Administrative Staff

These users will manage financial transactions, approve expenditures, generate reports, monitor budgets, and administer the system through the web application.

Barangay Residents

Residents will use the mobile application to view approved financial reports, monitor project budgets, receive announcements, and verify the authenticity of financial documents through QR code scanning.

Secondary Users

- Municipal Government Officials
- Commission on Audit (COA) Representatives
- Local Government Auditors
- Researchers and authorized government agencies

VI. PROPOSED TECHNOLOGIES

Web Application

- PHP
- JavaScript
- HTML5
- CSS3
- Bootstrap 5

Mobile Application

- Flutter (Dart)
- RESTful API (PHP)

Database

- MySQL

Blockchain Technologies

- Ganache (Private Blockchain)
- Solidity Smart Contracts
- Web3.js
- SHA-256 Cryptographic Hashing
- QR Code Generator

Development Tools

- Visual Studio Code
- Git & GitHub
- XAMPP

VII. INNOVATION

The proposed system introduces a web and mobile-based financial transparency platform that allows barangay officials to digitally manage financial transactions while providing residents with real-time access to approved financial reports through a mobile application.

Unlike traditional financial management systems, Aninag integrates blockchain-based verification by storing cryptographic hashes of finalized financial records in a private blockchain network. This ensures that official financial documents can be verified and that unauthorized modifications can be detected.

The integration of QR code verification, blockchain security, financial analytics, audit logs, and a public transparency portal provides a modern solution that promotes accountability, discourages corruption, and strengthens public trust in barangay governance.

VIII. FEASIBILITY

Technical Feasibility

The proposed system will use widely available and open-source technologies such as PHP, JavaScript, Flutter, MySQL, and Ganache. These technologies are suitable for undergraduate development projects and can be implemented using standard computer hardware.

Operational Feasibility

Barangay officials already perform financial recording and reporting as part of their daily responsibilities. The proposed system digitizes these existing processes, making adoption easier with minimal training. Residents can conveniently monitor approved financial reports through the mobile application.

Economic Feasibility

The project is cost-effective because it uses open-source software and can be developed using existing computer equipment. Deployment can be performed using affordable web hosting services, with operational costs limited to hosting, maintenance, and domain registration.

Schedule Feasibility

The proposed system can be completed within one academic term. Development will include requirements gathering, system design, implementation, testing, evaluation, and documentation using an incremental development approach.

IX. SCOPE AND LIMITATIONS

Scope

The proposed study focuses on the design and development of a web and mobile-based barangay transparency fund management system. The web application will be used by authorized barangay officials to record financial transactions, manage budgets, generate reports, monitor expenditures, and administer the system. The mobile application will enable residents to view approved financial reports, monitor project expenditures, receive announcements, and verify official financial documents using QR codes. The system also includes blockchain-based financial record verification, audit logging, and graphical analytics.

Limitations

The system is intended only for barangay financial transparency and does not replace official accounting systems required by national government agencies. Blockchain implementation is limited to document verification through a private blockchain environment and does not involve cryptocurrency transactions or public blockchain networks. Only approved financial reports and summaries will be accessible to residents, while confidential financial information will remain available only to authorized barangay personnel. The system requires an internet connection and is not integrated with banking systems or other external government financial platforms.

X. EXPECTED OUTPUT

Describe deliverables such as prototype, dashboard, reports, or mobile app.